

Changing Paradigms of Feline Idiopathic Cystitis

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Disorders of the feline lower urinary tract that are characterized by periuria (urination in inappropriate locations), hematuria, dysuria, pollakiuria, and/or urinary obstruction are common but incompletely understood problems. Research over the past 50 years has reaffirmed the notion that cats are similar to other species, including dogs and humans with regard to the fact that lower urinary tract signs may result from a variety of diverse disorders including: bacterial, fungal, and parasitic urinary tract infections; different types of uroliths and urethral plugs; neoplasia, congenital or acquired anatomic or morphologic abnormalities; and traumatic, neurogenic, or iatrogenic causes.^{1–4} However, these same studies also serve to emphasize that some young to middle-aged cats are uniquely predisposed to lower urinary tract signs for which precise causes have not been identified. These cats are classified as having idiopathic lower urinary tract disease or idiopathic cystitis. The disorder is believed to affect one quarter to one half a million cats in the United States annually.⁵ As of yet, there is no consistent specific diagnostic marker and there is no consistently effective method of therapy.⁶

The term “paradigm” refers to a generally accepted theoretical model that explains complex ideas.⁷ Over that past several decades, the prevailing paradigms regarding the causes of cystitis have changed because of: the influence of contemporary studies of the biological behavior of naturally occurring idiopathic cystitis; laboratory models of experimentally induced lower urinary tract diseases; and extrapolation from studies in other species.⁸ It is clear, however, that no single model explains all the biological variability observed in cats with idiopathic cystitis. In many respects, feline idiopathic cystitis is a conundrum analogous to an idiopathic lower urinary tract disorder of humans called interstitial cystitis.⁹ The extent to which these two conditions share

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etiopathogenic mechanisms has not yet been precisely defined; however, the diversity in their biological behavior and responses to therapeutic interventions, and their association with comorbid conditions, suggest that these disorders may represent syndromes resulting from a variety of separate underlying but potentially interrelated mechanisms rather than one disease with a single uniform pathogenesis.^{8,10,11} Design of effective therapeutic and prevention strategies is ultimately dependent on identifying the specific etiologic factors responsible for feline idiopathic cystitis.

BIOLOGICAL BEHAVIOR

Periuria, pollakiuria, stranguria, and gross hematuria are the most common clinical signs observed in cats with nonobstructive idiopathic cystitis and may precede the obstructive form of the disorder. Remarkably, these clinical signs subside within 5–7 days without therapy in up to 92% of cats with acute nonobstructive idiopathic cystitis.^{12–15} Signs may recur after variable periods of time and again subside without treatment. However, recurrences of signs have been reported to affect 39% to 65% of cats with acute idiopathic cystitis within one to two years after the initial episode.^{12,14,16,17} In one study, survival analysis revealed that an increased number of prior episodes of lower urinary tract signs was associated with a significantly higher risk of recurrence of clinical signs, while increased age was associated with a significantly lower risk of recurrence of clinical signs.¹⁴ After controlling for age, number of prior episodes of lower urinary tract signs, and diet moisture content, the relative risk that an affected cat will have a recurrent episode of lower urinary tract signs in a 2-year period was approximately 65%. These data and our clinical observations suggest that recurrent episodes of acute idiopathic cystitis decrease in frequency and severity as the cats become older.^{13,14,18} Though recurrent clinical signs in patients with idiopathic cystitis are often assumed to be a recurrent episode of the original disease, recurrent signs may also be the result of a delayed manifestation of the original disease (for example, spontaneous or iatrogenic urethral stricture), or a different lower urinary tract disease associated with similar clinical signs (such as urolithiasis).

The authors of this article and others have also encountered a small subset of cats with idiopathic cystitis in which clinical signs persist for weeks to months or which are frequently recurrent.^{5,19,20} These cats are classified as having chronic idiopathic cystitis. In the authors' experience, less than 15% of cats that initially presented with acute idiopathic cystitis develop this form of the disease. (Kruger JM, unpublished data, 2003) It has also been the authors' experience that some cats with chronic or frequently recurrent forms of idiopathic cystitis experience spontaneous remission of their clinical signs. Whether chronic idiopathic cystitis represents one extreme in the spectrum of clinical manifestations associated with similar etiologic factors, or whether it represents an entirely different mechanism of disease than that associated with acute self-limiting idiopathic disease is as yet unknown. To date, it appears that the vast majority of studies investigating potential causes of idiopathic cystitis involved cats affected with chronic forms of the disease.

BACTERIAL UROPATHOGEN PARADIGM

In the 1960s and 1970s, the prevailing theory of the cause of lower urinary tract signs was bacterial urinary tract infection (UTI).^{21–23} Before 1970, bacterial UTI was one of the most commonly recognized and treated disorders on the lower urinary tract in dogs.²⁴ Because of the paucity of reproducible evidence about naturally occurring diseases in cats at that time, it was common to extrapolate and apply information derived in other species to cats. The notion that bacteria were commonly involved in feline lower

urinary tract diseases was reinforced by erroneous identification and interpretation of bacterial-look-alikes (aka pseudobacteria) found in urine sediment as bacterial pathogens. In a recent study in the authors' laboratory, up to 40% of unstained feline urine sediments classified as positive for bacteriuria by light microscopy were found to be false positives when compared with results of quantitative culture of urine for bacteria.²⁵ Microscopic examination of unstained urine sediment was associated with only an 11% positive predictive value (ie, the proportion of cats with a positive test that had confirmed bacterial UTI). Staining of urine sediment with a modified Wright-stain procedure (Diff Quik®) reduced the frequency of false positives to approximately 2%.^{25,26}

In addition to reliance on faulty results of urinalysis, false-positive bacterial culture results associated with improper urine collection techniques and failure to quantify the number of bacterial isolates were pervasive. As a result, bacterial contaminants were misidentified as bacterial pathogens. The interpretation that bacteria were an important cause of lower urinary tract signs in cats was further enhanced by the observation that clinical signs often subsided in association with antimicrobial therapy. Based on the observation that urinary host defense mechanisms in cats are of such design that cats are innately resistant to bacterial infections,^{27,28} it is certain that many affected cats with presumed bacterial UTI appeared to respond favorably to antimicrobics because they had self-limiting forms of idiopathic cystitis.

Although it has been well established that aerobic bacteria account for only 1–3% of bacterial UTI in young to middle-aged cats, empirically selected antimicrobics remain as the most common form of therapy prescribed for cats with hematuria, dysuria, polakiuria, and periuria.^{1–4,28–31} Even though the uselessness of antimicrobial agents in the treatment of abacteriuric cats with lower urinary tract disease also has been well documented, the bacterial UTI paradigm persists.¹²

VIRAL UROPATHOGEN PARADIGM

One attractive hypothesis advanced in the late 1960s and early 1970s implicated viruses as causative agents in the etiopathogenesis of some forms of naturally occurring feline lower urinary tract diseases.^{31–33} This hypothesis was supported by the isolation of a gamma herpesvirus (aka bovine herpesvirus type 4), retroviruses (feline foamy virus [FFV]), and a calicivirus (feline calicivirus [FCV]) from urine and tissues obtained from cats affected with lower urinary tract disease.^{34–38} Similarly, a number of viral agents including adenoviruses, polyomaviruses, herpesviruses, influenza A, and human immunodeficiency virus, have been associated with hemorrhagic cystitis and other lower urinary tract symptoms in humans.^{39,40}

Feline Calicivirus

The first indirect evidence supporting the hypothesis of a viral etiology in feline lower urinary tract disorders was reported in 1969. Investigators at Cornell University produced urethral obstruction in male cats by urinary bladder inoculation with centrifuged bacteriologically sterile urine obtained from male cats with naturally occurring urethral obstruction.⁴¹ Subsequent isolation of FCV from a Manx cat with spontaneous urethral obstruction, and induction of obstructive uropathy in conventionally reared cats by urinary bladder inoculation with this virus, supported the concept of a viral etiology for feline lower urinary tract disease.^{41,42} In a study of induced FCV urinary tract infection in conventionally reared cats, 80% of cats developed urethral obstruction following urinary bladder, aerosol, or contact exposure.⁴² However, failure to reisolate FCV from urine after the fourth day of inoculation, lack of a significant serum neutralizing antibody response, and isolation of an additional virus (feline foamy virus)

from all obstructed experimental cats, prompted the Cornell investigators to hypothesize that FCV was not a primary causative agent.³² The notion was advanced that this virus incited other latent viruses present in the urinary tract to induce urethral obstruction. Results of a subsequent study comparing the affects of induced urinary tract infections with FCV alone or in combination with a gamma herpesvirus appeared to support the concept that FCV exacerbated the severity of disease in cats infected with gamma herpesvirus.^{31,32}

The questionable nature of the etiologic role of FCV followed reports of other investigators who were unable to obtain reproducible evidence that caliciviruses were associated with naturally occurring lower urinary tract diseases of cats.^{3,29,33,36,37} In retrospect, however, inability of investigators to detect FCV in cats with lower urinary tract disorders in the 1970s must be viewed with caution. Their inability to detect FCV appeared to be confounded by improper selection of cases (ie, improper inclusion and exclusion criteria were used to select cats for study), the innately virucidal nature of feline urine, and use of insensitive or inappropriate virus detection methods.^{33,43,44} The point to keep in mind is that absence of evidence cannot always be interpreted as evidence of absence.

Feline Foamy (Syncytium-Forming) Virus

Shortly after the isolation of FCV, the Cornell group isolated the syncytium-forming retrovirus FFV (aka feline syncytium-forming virus) from several cats with lower urinary tract signs.^{35,36,38} Unfortunately, there have been limited studies designed to investigate the etiopathogenic role of FFV in cats with lower urinary tract disease.³³ In one study, FCV-induced urethral obstruction in conventionally reared cats was consistently associated with isolation of FFV.⁴² In other studies, lower urinary tract clinical signs were not observed in a small number of cats following intravenous, intraperitoneal, intramuscular, intra-articular, or subcutaneous inoculation of FFV.⁴⁵⁻⁴⁸

In contrast to other viruses, FFV has been isolated from urine, urinary-tract tissues, and other tissues obtained from a large number of cats with lower urinary tract disease.³³⁻³⁸ Additionally, FFV antibodies have been detected in serum samples obtained from cats with naturally occurring lower urinary tract disease.^{33,38} FFV antibodies have also been detected in a large number of clinically normal cats.⁴⁸ The relative ease and frequency with which FFV has been isolated from cats with lower urinary tract signs and the prevalence of FFV antibodies suggests that they may play a potential role in the pathogenesis of the disease. However, it is difficult to assess the relative importance of FFV on the basis of available experimental and clinical evidence. Further investigations are necessary to determine the role (if any) of FFV virus in feline idiopathic cystitis.

Gamma Herpesvirus

In addition to FCV and FFV, investigators at Cornell University isolated a gamma herpesvirus (aka bovine herpesvirus type 4 [BHV-4]) from pooled kidney organ explants obtained from a litter of normal kittens, a kitten with upper respiratory disease, and a kitten with concurrent upper respiratory disease and urethral obstructions.^{35,49} Results of studies designed to induce urethral obstruction in specific-pathogen-free (SPF) cats by urinary bladder inoculation of FCV alone, herpesvirus alone, or herpesvirus in combination with FCV, led to the hypothesis that gamma herpesvirus was a primary causative agent in the etiopathogenesis of naturally occurring feline lower urinary tract diseases.^{31,32}

In an effort to substantiate Cornell's studies and to further characterize the clinicopathologic manifestations of gamma herpesvirus-induced lower urinary tract disease,

the authors inoculated the feline isolate of BHV-4 into the urinary bladders of SPF male and female cats.⁵⁰ Results of these studies indicated that BHV-4 was capable of establishing persistent low-grade or latent urinary tract infections in male and female SPF cats. However, in contrast to previous studies, lower urinary tract signs were uncommonly associated with persistent herpes virus urinary tract infections.

Serologic methods also have been used to evaluate the role of gamma herpes virus in feline lower urinary tract disorders. In a prospective study, sera obtained from 167 normal cats and cats with obstructive or nonobstructive idiopathic cystitis, urolithiasis, or bacterial UTI, were evaluated for BHV-4 using an indirect fluorescent antibody test.³ Antibodies against BHV-4 were detected in sera obtained from 31% of cats with lower urinary tract diseases, and from 23% of normal control cats. However, positive BHV-4 antibody titers were not associated with clinical signs or abnormal laboratory findings. These observations and the inability of other investigators to isolate gamma herpes viruses from cats with naturally occurring lower urinary tract disorders or to induce lower urinary tract signs in healthy cats precludes assigning BHV-4 a primary role in feline idiopathic cystitis. Considering the relatively high prevalence of BHV-4 antibodies in the general feline population, additional studies defining the pathogenic role of gamma herpesvirus infections of cats are warranted.⁵¹

VESICOURACHAL DIVERTICULUM PARADIGM

In the late 1970s and early 1980s, vesicourachal diverticula were cited as playing an etiopathogenic role in cats with lower urinary tract signs.^{52,53} A vesicourachal diverticulum is a common congenital anomaly of the urinary bladder that occurs when the portion of the urachus (ie, a fetal conduit that allows passage of urine from the bladder to the placenta during development of the fetus) located at the bladder vertex does not completely close.⁵⁴ The result is a blind diverticulum of variable size located in the bladder vertex. Congenital microscopic diverticula are remnant urachal canals lined with urothelium that may persist from the level of the submucosa to the serosa.⁵⁵ In a study of 80 feline urinary bladders, microscopic urachal diverticula were detected in more than 40%.⁵³ Microscopic remnants persisting in the urinary bladder vertex after birth are clinically silent.

Acquired macroscopic diverticula may develop at the bladder vertex of cats with microscopic urachal remnants as sequelae to concurrent but unrelated acquired lower urinary tract diseases (eg, bacterial infections, urolithiasis, crystalline-matrix urethral plugs, idiopathic disease, etc.).⁵⁶ Presumably, urethral obstruction and/or detrusor hyperactivity induced by inflammation result in increased intraluminal pressure and subsequent enlargement of microscopic diverticula.⁵⁵ Congenital macroscopic diverticula are most likely caused by impaired urine outflow. They develop before or soon after birth and, unlike most acquired diverticula that spontaneously regress, congenital macroscopic diverticula persist indefinitely.

In the early 1990s, many surgeons recommended treatment of vesicourachal diverticula by surgical extirpation.^{57,58} The observation that clinical signs coincidentally subsided after diverticulectomy and a lack of studies of the biological behavior of macroscopic diverticula without surgery reinforced the opinion that diverticulectomy was an effective surgical method to eliminate lower urinary tract signs associated with this disorder. Contrary to the prevailing opinion of the time, however, results of clinical studies revealed that diverticulectomy was not required in most cats because adult-onset acquired macroscopic vesicourachal diverticula typically resolved within 2 to 3 weeks after amelioration of clinical signs of lower tract disease.⁵⁶ After being resolved, vesicourachal diverticula did not recur in association with subsequent

episodes of lower urinary tract disease. These observations provide convincing evidence that most vesicourachal diverticula associated with clinical signs are the result and not the cause of idiopathic cystitis.

STRUVITE CRYSTALLURIA PARADIGM

In the early 1970s, the association between dry diets and lower urinary tract signs in cats became a topic of intense discussion. Also in the early 1970s and continuing for the next decade, several groups of investigators experimentally produced magnesium hydrogen phosphate and then magnesium ammonium phosphate uroliths in normal cats by addition of various types of magnesium salts to their diets.⁵⁹ The cats developed typical signs of lower urinary tract disease including urethral obstruction. However, this model did not result in production of struvite-matrix urethral plugs commonly encountered in cats with naturally occurring urethral obstruction.⁶⁰ Nonetheless, as the viral paradigm lost popularity in the late 1970s, the general consensus of many investigators and clinicians was that consumption of dry diets with excessive magnesium was an important primary cause of feline lower urinary tract disease.⁸

Beginning in 1983, reports from the Minnesota Urolith Center of medical dissolution of naturally occurring sterile struvite urocystoliths with modified diets corroborated the importance of dietary risk and protective factors in the etiology of sterile struvite urolithiasis.^{61,62} In 1985, investigators at the University of California performed studies in normal cats using diets containing alkalinizing and acidifying salts of magnesium.⁶³ Results of these studies shifted the focus of attention from dietary magnesium to alkaline urine pH as a primary factor involved in the development of struvite crystalluria. Results of these diet-related studies also supported the paradigm in that era that many cases of feline lower urinary tract disease were caused by struvite uroliths and urethral plugs. The pet food industry responded to these recommendations by placing increased emphasis on development of urine acidifying diets modified to dissolve and prevent struvite crystalluria. Many adult feline maintenance diets were modified to minimize struvite crystalluria.⁶⁴

There is abundant evidence that, in past years, sterile struvite uroliths and struvite urethral plugs played a prominent causative role in a substantial number of cats with naturally occurring obstructive and nonobstructive lower urinary tract disease. In addition, dietary factors contributed both to the cause and the treatment and prevention of struvite related lower urinary tract diseases. As a result of dietary modifications by numerous pet food manufacturers in the mid-1980s, the prevalence of struvite uroliths and struvite urethral plugs began to decline (**Fig. 1**). Although struvite continued to be the primary component in urethral plugs (**Fig. 2**), the frequency with which they formed dramatically declined as evidenced by a parallel decline in the number of perineal urethrostomies performed to minimize recurrent urethral obstruction.⁶⁵ Unexpectedly, the decline in struvite related urolithiasis was associated with a concomitant rise in calcium oxalate urolithiasis, while, unexplainably, struvite remained as the primary component of urethral plugs (see **Figs. 1** and **2**).

Did the results of these extensive investigations resolve the search for the primary causes of lower urinary tract disease? The observation that signs of hematuria and dysuria occurred in more than 50% of cats with idiopathic cystitis in the absence of any type of crystalluria indicated that other as yet unidentified etiologic factors were also involved.^{1-3,14,18}

DYSFUNCTIONAL UROTHELIAL BARRIER PARADIGM

Over the past two decades, urothelial barrier dysfunction has been a pervasive paradigm of causation of human interstitial cystitis.^{10,66,67} Likewise, feline studies in the

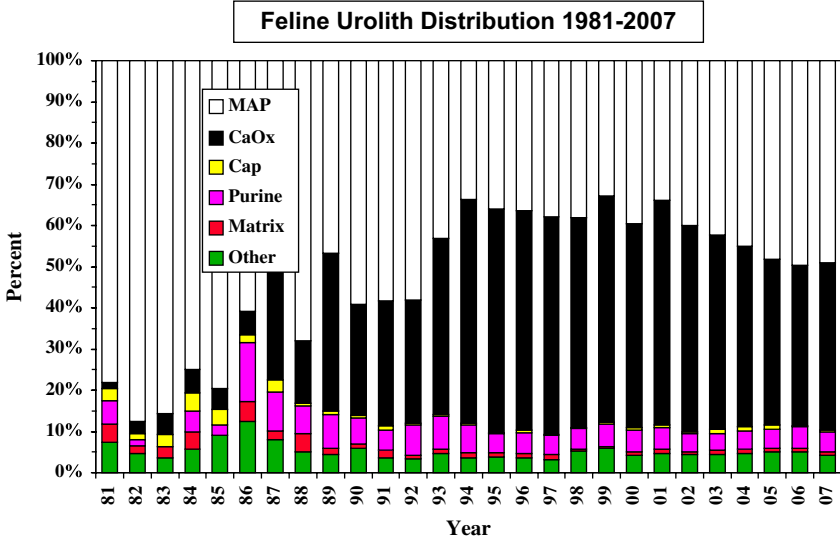


Fig.1. Changing trends in mineral composition of feline uroliths submitted to the Minnesota Urolith Center from 1981 to 2007.

mid- to late-1990s supported the concept that urothelial barrier failure may be involved in the pathogenesis of idiopathic cystitis.⁶⁸⁻⁷⁰ Bladder urothelium is an important host defense mechanism because it serves as a barrier that prevents entry of uropathogens into deeper structures. Urothelium also selectively controls passage

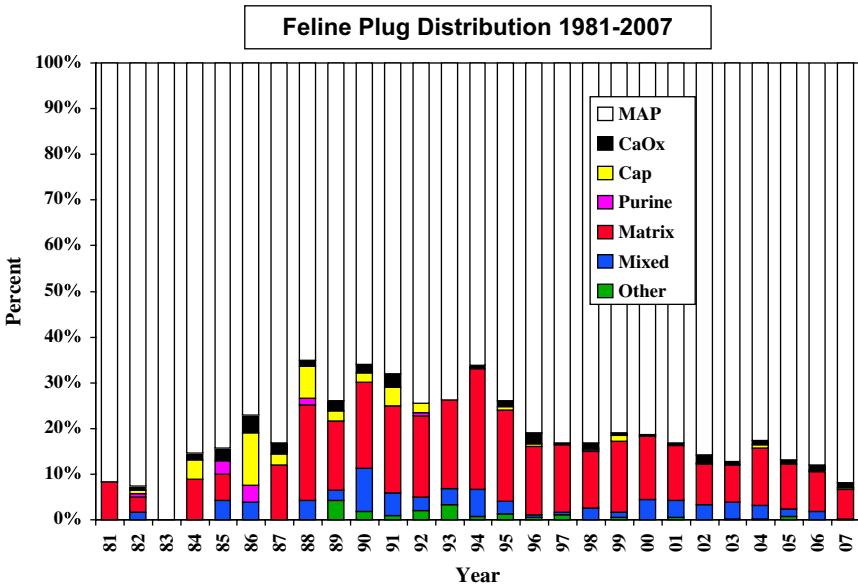


Fig. 2. Changing trends in mineral composition of feline urethral plugs submitted to the Minnesota Urolith Center from 1981 to 2007.

of water, ions, macromolecules, and other solutes across the mucosal surface into underlying tissues.^{66,71} Urothelial barrier function largely depends on: 1) specialized high-resistance tight junctions between apical membranes of adjacent urothelial cells; 2) the unique lipid and protein composition of urothelial cell membranes; and 3) a layer of glycosaminoglycans (GAG) located on the luminal surface of urothelial cells.^{71,72} There is disagreement as to the relative contributions of the GAG layer and urothelium as the barrier to urine ions, macromolecules, toxins, and microorganisms.^{66,72} Regardless, any disease process that: 1) directly injures urothelium; 2) alters structural or functional characteristics of urothelial GAGs or apical tight junctions; and/or 3) disrupts active transport mechanisms may result in loss of barrier function.⁷²⁻⁷⁴ These changes may allow translocation of toxic substances into underlying tissues, resulting in submucosal edema, hemorrhage, neovascularization, mastocytosis, mononuclear inflammatory cell infiltration, fibrosis, and clinical signs of frequency, urgency, and pain.^{70,73,75-80}

Defective Glycosaminoglycan Layer

Transitional epithelium of the urinary bladder is covered by a glycocalyx composed of hydrated glycoconjugates, including glycoproteins and glycosaminoglycans (GAGs).⁶⁶ Urothelial GAGs minimize adherence of microorganisms and crystals to the bladder urothelium and also limit movement of urine proteins and other ionic and nonionic solutes from the bladder lumen into surrounding tissues.^{66,67,81} Quantitative or qualitative defects in surface GAGs and subsequent increased urothelial permeability have been hypothesized to be a causative factor in the pathogenesis of feline idiopathic cystitis and human interstitial cystitis.^{20,67,81,82} Chronic exposure of bladder wall tissues to urine constituents could result in sensory afferent nerve stimulation, mast cell activation, and/or induction of immune-mediated or neurogenic inflammatory responses.¹⁰ Compared to normal cats, some cats with idiopathic cystitis appear to have increased urinary bladder permeability to salicylates, urea, and fluorescein, decreased surface GAG expression, and decreased total urinary GAG excretion.^{68-70,83,84} Investigators have identified similar abnormalities in humans with interstitial cystitis.^{81,85,86} However, others have failed to detect altered urothelial GAG layers, decreased urine GAG excretion, or enhanced urothelial permeability in humans with interstitial cystitis.⁸⁷⁻⁹⁰ The controversy surrounding the role of GAGs in humans with interstitial cystitis has been confounded by the fact that increased urothelial permeability has been associated with other non-interstitial cystitis lower urinary tract disorders (eg, urolithiasis, chemical cystitis, and bladder overdistention).^{10,91,92} Nevertheless, replacement of the urothelial GAG layer with oral or intravesicular administration of pentosan polysulfate sodium (an exogenous semisynthetic GAG analog) appears to significantly reduce the severity of symptoms in some human patients with interstitial cystitis.⁹³⁻⁹⁵ In cats with idiopathic cystitis, however, results of a randomized placebo-controlled clinical trial of oral glucosamine (a GAG precursor) did not reveal any significant difference between the severity of clinical signs in cats treated with glucosamine and those treated with a placebo.¹⁶ Further characterization of structural or functional alterations of the urothelial GAG layer in cats with and without idiopathic cystitis are essential to determine whether these abnormalities are a specific cause of idiopathic disease or a nonspecific effect occurring secondary to other etiologic mechanisms.

Defective Urothelial Proliferation or Differentiation

Urothelium is stratified and consists of at least three layers: a basal cell layer attached to the basement membrane; an intermediate cell layer; and a superficial apical layer

composed of large hexagonal cells (so-called “umbrella cells”).^{71,72} The two deeper layers of urothelial cells serve as progenitors for the overlying superficial layer. Superficial urothelial cells are responsible for maintaining the bladder permeability barrier and possess unique features including: 1) specialized apical membrane proteins (called uroplakins); 2) high resistance tight junctions between cells; and 3) an active trafficking mechanism designed to insert cytoplasmic vesicles into the apical membrane to accommodate bladder filling.⁷² Primary responses of urothelium to injury are desquamation (exfoliation of viable cells), necrosis, and/or apoptosis.^{74,96,97} In healthy animals, rapid replacement of exfoliated or injured superficial cells and restoration of urothelial tight junction integrity prevents long-term urothelial barrier malfunction.⁹⁸ However, urothelial injury in the face of any disease process that alters normal urothelial proliferation and differentiation could profoundly affect urothelial barrier function.

Urothelial ulceration, tearing, and thinning are common abnormalities identified in biopsy specimens from humans with interstitial cystitis.^{99–101} These observations led investigators to hypothesize that impaired urothelial proliferation may be involved in the pathogenesis of human interstitial cystitis.¹⁰² Subsequent identification of abnormal expression of protein biomarkers associated with urothelial growth and differentiation (eg, uroplakins, E-cadherin, ZO-1, glycoproteins, GAGs, and keratins) in biopsy specimens from humans with interstitial cystitis further supported this hypotheses.^{66,101,103–105} Also, a novel low molecular weight sialoglycopeptide called antiproliferative factor (APF), which inhibited proliferation of normal bladder urothelial cells in vitro, was identified in the urine of humans with interstitial cystitis.¹⁰⁶ Subsequent in vitro studies revealed that APF suppressed urothelial cell proliferation, increased transcellular permeability, inhibited tight junction protein expression, and reduced growth factor production.^{107–109} Unfortunately, detection of urine APF activity as a marker of interstitial cystitis is dependent on bioassays using human urothelial cell cultures. There are no available commercial assays for detection of APF in human or feline urine at this time.¹⁰⁶

Urothelial ulceration, erosion, and thinning are common light microscopic features of chronic forms of feline idiopathic cystitis.^{70,77,80} Despite histopathologic similarities to human interstitial cystitis, there have been relatively few studies evaluating protein biomarkers of urothelial growth and urothelial differentiation in cats with acute or chronic idiopathic cystitis. In a study of three cats with chronic idiopathic cystitis, urothelial cells underlying areas of mucosal erosion expressed abundant levels of the superficial urothelial cell differentiation marker AE-31.⁷⁰ However, distribution of AE-31 antigen was altered with staining localized to the perinuclear region of cells rather than at its normal apical membrane position. Likewise, results of a study of 40 cats with chronic idiopathic cystitis revealed that the magnitude of immuno-histochemical staining for uroplakin was significantly less in affected cats compared with normal cats but not significantly different compared with cats with urolithiasis.⁸⁰ It is unclear whether alterations in urothelial markers in cats with idiopathic cystitis represent inherent dysfunction of the urothelium or a secondary response to urothelial injury. Nevertheless, systematic evaluations of protein biomarkers of urothelial differentiation and their role in the pathogenesis of feline idiopathic cystitis as well as their potential as noninvasive diagnostic markers of disease warrant further investigations.

NEUROGENIC INFLAMMATION PARADIGM

Neurogenic inflammation is a process initiated by excitation of small c-fiber sensory afferent neurons and mediated by neuropeptides (eg, substance P, neurokinin, and

calcitonin gene-related peptide) released from stimulated nerves.^{10,110} Structural or functional defects in the urothelial barrier could permit hydrogen, calcium, and potassium ions, or other urine constituents to come into contact with sensory afferent neurons innervating the urothelium. Alternatively, injury-induced alterations in urothelial release of chemical signaling molecules (eg, ATP, nitric oxide, acetylcholine, substance P, and prostaglandins) may activate sensory afferent neurons and mast cells.^{10,73,111} Neurogenic inflammation may also be triggered by histamine and other mediators released by independently activated mast cells in close proximity to neuropeptide-containing sensory neurons.^{10,110} Interaction of neuropeptides with tissue receptors results in vasodilation, increased vascular permeability, increased epithelial permeability, increased leukocyte migration, and mast cell activation.^{112–115} It is apparent that the combined effects of neuropeptides and mast cell mediators may induce a wide range of biological effects culminating in pain, inflammation, tissue injury, and fibrosis.^{10,116} Observations of increased numbers of substance P–containing nerve fibers and their close anatomic association with mast cells in urinary bladders of humans with interstitial cystitis have led to the hypothesis that neurogenic inflammation (and subsequent mast cell activation) represents the common pathway in a multifactorial pathogenesis of human interstitial cystitis.^{10,111,117–121}

Similarly, increased numbers of substance P–containing sensory afferent neurons and high affinity substance P receptors have been observed in the bladder submucosa of cats afflicted with chronic idiopathic cystitis.¹²² These observations suggest that increased numbers of neuropeptide containing neurons and increased expression of high affinity substance P receptors are associated with inflammation of feline urinary bladders.⁸² However, the question of whether these phenomena are a cause or an effect of inflammation and whether they have a role in the etiopathogenesis of feline idiopathic cystitis requires additional investigation.

MAST CELL (NEUROIMMUNE) PARADIGM

When compared with normal cats, increased numbers of mast cells have been identified within the bladder submucosa of some, but not all, cats with idiopathic cystitis.^{75,76,80,123} Likewise, mast cell infiltrates are often seen in bladder biopsies obtained from humans with interstitial cystitis.^{11,120,124,125} Mast cells secrete a variety of preformed biologically active molecules (histamine, heparin, serotonin, kinins, proteases, phospholipases, chemotactic factors, cytokines, and vasoactive intestinal peptide) as well as molecules synthesized de novo (interleukin-6, leukotrienes, platelet activating factor, prostaglandins, thromboxanes, nitric oxide, tumor necrosis factor- α).^{111,120,125} Mast cell secretion can be triggered by anaphylatoxins, antigens, bradykinin, cytokines/lymphokines, hormones, IgE, neurotransmitters, neuropeptides, bacterial toxins, viruses, drugs, and stress.^{120,125–127} In addition, release of cytokines from damaged or dysfunctional urothelium is a potent mast cell activator.¹²⁰ Products of activated mast cells could be responsible for the inflammation, pain, vasodilation, fibrosis, and smooth muscle contraction associated with human interstitial cystitis and feline idiopathic cystitis.^{10,111,120,123,125} Histamine and its metabolites have been found in higher concentrations in the urine of human interstitial cystitis patients and the cystoscopic effluent of cats with idiopathic cystitis.^{119,123,128,129}

Despite these observations, the role of mast cells in human interstitial cystitis and feline idiopathic cystitis are unknown. Excessive numbers of mast cells have been observed in bladder biopsies obtained from humans with bladder diseases other than interstitial cystitis, such as, bacterial UTI, neoplasia, outflow obstruction, and stress incontinence.^{121,124,125,130} Likewise, the number

of mast cells immuno-histochemically stained with c-kit antibody in bladder biopsies obtained from cats with chronic idiopathic cystitis were not significantly different from the number observed in biopsies obtained from cats with urolithiasis.⁸⁰ These observations suggest that urinary bladder mastocytosis may not be specific for either human interstitial cystitis or feline idiopathic cystitis. However, electron microscopic studies indicate that the majority of urinary bladder mast cells in humans with interstitial cystitis are activated and are often located in close proximity to neuropeptide-containing sensory nerves. In contrast, mast cell activation is not a feature of lower urinary tract disorders in humans from which interstitial cystitis has been excluded.^{117–119,121} The close biochemical and anatomic relationship between urinary bladder mast cells and neuropeptide-containing neurons have led to the hypothesis that neurohormonal triggering of mast cell activation and subsequent release of mast cell mediators play a central role in a multifactorial pathogenesis of human interstitial cystitis.^{10,111,118,121} Similar ultrastructural studies of mast cells and their spatial relationship to sensory nerves in cats with idiopathic cystitis have not been performed.

SYSTEMIC PSYCHONEUROENDOCRINE DISEASE PARADIGM

One of the major paradigms shifts over the past decade implicates psychoneuroendocrine dysfunction as a causative factor in the pathogenesis of feline idiopathic cystitis.^{20,131,132} Clinical observations suggest that chronic stress induced by environmental, psychologic, physiologic, or comorbid pathologic conditions may play a role in precipitating or exacerbating signs associated with chronic forms of idiopathic cystitis.^{133–136} Stress may be defined as any physical, chemical, or emotional force that disturbs or threatens homeostasis, and the accompanying adaptive responses that attempt to restore homeostasis.¹³⁷ Although generally beneficial, adaptive responses themselves appear to serve as stressors capable of inducing pathologic changes. Stress has been implicated as a pathologic factor in disease in general, and in autoimmune and neuroinflammatory diseases in particular. It is interesting that stressful events such as earthquakes, seasonal weather changes, moves to a new home, major holidays, and diet changes have been associated with recurrent episodes of lower urinary tract signs in cats.^{132,135,136} Other potentially stressful circumstances for cats, such as living in multicat households and inter-cat conflicts, have also been associated with an increased risk of disorders of the lower urinary tract.^{133,134,136} Similarly, in humans with interstitial cystitis, stress appears to be a trigger for precipitation or exacerbation of clinical signs.¹³⁸

The association of idiopathic cystitis with environmental, psychologic, physiologic, and pathologic stressors and the identification of multiple abnormalities of the nervous and endocrine systems in affected cats has led to hypothesis that systemic psychoneuroendocrine factors may be involved in the pathogenesis of the disease.^{133,139–141} In this paradigm, the urinary bladder is considered to be more likely a victim rather than a perpetrator.^{20,131,132} Acute or chronic stress from any source may increase activity of tyrosine hydroxylase (the rate limiting enzyme catalyzing norepinephrine biosynthesis in the pontine locus coeruleus and lead to increased sympathetic autonomic outflow.¹⁴² Observations of increased tyrosine hydroxylase immunoreactivity¹⁴⁰ and plasma norepinephrine concentrations,^{84,139} and decreased functional sensitivity of alpha-2-adrenoceptors¹⁴³ in cats with chronic idiopathic cystitis are consistent with increased sympathetic drive in affected cats. Activation of the locus coeruleus/noradrenergic system is an important physiologic response to stress; however, it has not

been determined whether this response is a cause or consequence of bladder abnormalities observed in cats with chronic idiopathic cystitis.^{139,140}

In addition to the locus coeruleus/noradrenergic system, exposure to stress simultaneously activates the hypothalamic-pituitary-adrenal axis (HPAA).^{144,145} Stress induced release of glucocorticoids appears to restrain catecholamine synthesis, release, reuptake, and metabolism in the locus coeruleus/noradrenergic system.^{145–148} In two small studies of cats with chronic idiopathic cystitis, significant abnormalities of the HPAA were not identified in affected cats, despite having significantly increased plasma norepinephrine concentrations.^{84,139} However, in a larger study, significantly decreased plasma cortisol responses to exogenous ACTH and reduction in the size of adrenal glands were observed in cats with chronic idiopathic cystitis compared with healthy cats.¹⁴¹ Taken together, these observations were interpreted to be indicative of dissociation between the responses of the sympathetic nervous system and hypothalamic-pituitary-adrenal axis to stress in cats with chronic idiopathic cystitis.^{84,131,141,144} Furthermore, it was hypothesized that enhanced central noradrenergic drive in the face of inadequate adrenocortical restraint may have a role in the disease process.^{131,144}

Specific mechanisms by which systemic psychoneuroendocrine disorders in cats may precipitate clinical signs referable to the lower urinary tract apparently have not been identified. Proposed mechanisms include stress-induced central dysregulation of autonomic neurons regulating bladder contraction and stress-induced direct effects on urothelium.^{84,144} In cats, the pontine micturition center (aka Barrington's nucleus) refers to an area in the dorsolateral pons and is located adjacent to the locus coeruleus.^{149–151} The diversity of afferent and efferent projections to and from Barrington's nucleus indicates that the micturition center functions to integrate pelvic visceral function with behavior. It also provides a balance of inhibitory and excitatory inputs to the sacral parasympathetic neurons that control bladder contractions.¹⁵¹ The presence of tyrosine hydroxylase-immunoreactive fibers from the locus coeruleus and the high density of glucocorticoid receptors in Barrington's nucleus suggest that stress-induced activation of either the locus coeruleus/adrenergic system or the HPAA may influence micturition.^{151–153} Activation of the micturition center via the locus coeruleus/adrenergic system or HPAA could increase excitatory efferent output to pelvic parasympathetic neurons that regulate bladder contraction. Conversely, projections from Barrington's nucleus to the locus coeruleus can also serve as potent activators of the locus coeruleus/noradrenergic system.^{151,154} It is conceivable that lower urinary tract disorders that result in chronic stimulation of bladder sensory afferents to Barrington's nucleus may induce persistent activation of the locus coeruleus/noradrenergic system.

Alternatively, it has been hypothesized that physiologic responses to stressors may directly affect urothelial integrity.¹⁵⁵ In rodent models, environmental, psychologic, or pharmacologic stressors induced loss of urothelial tight junction integrity, increased paracellular permeability, activation of mast cells, and desquamation of viable superficial urothelial cells.^{97,98,127,155,156} In healthy animals, rapid replacement of exfoliated superficial cells and restoration of the urothelial tight junction integrity prevents long-term malfunction of the urothelial barrier.⁹⁸ However, long-term increases in the rate of desquamation could have profound effects on urothelial barrier integrity, especially in circumstances where normal urothelial proliferation is inhibited. Although similar studies have not been reported in cats, it is plausible that activation of the sympathetic nervous system and subsequent increases in norepinephrine could increase urothelial permeability by reducing tight junction integrity, thus permitting urine substances greater contact with bladder sensory afferent neurons.^{84,157}

Based on psychoneuroendocrine abnormalities identified in affected cats, multi-modal environmental modification (MEMO) strategies designed to minimize reactivity of the stress response system have been advocated as primary therapy for prevention and recurrence of feline idiopathic cystitis.^{131,157,158} In an uncontrolled study,¹⁵⁷ 46 client-owned cats living indoors and with a history of recurrent episodes of lower urinary tract signs were managed with individualized plans to reduce environmental and social stressors including changes in physical environment (n=16), increased owner-cat interaction (n=10), moist diet (n=14), enhanced litter box hygiene (n=12), efforts to reduce inter-cat conflict (n=3), and pharmacologic agents (amitriptyline n=7, clomipramine n=1, acepromazine n=2). During the 10-month follow-up period, there was a significant reduction in the frequency of lower urinary tract signs with approximately 70–75% of affected cats remaining symptom free. Unfortunately, the uncontrolled nature of the study and concurrent use of dietary and/or pharmacology interventions in some cats makes it impossible to determine the relative clinical benefits of MEMO. Nevertheless, systematic evaluation of stress factors and their influence on the incidence and pathogenesis of feline idiopathic cystitis and the efficacy of therapeutic strategies designed to mitigate stress in affected cats warrant further investigations.

VIRAL UROPATHOGEN PARADIGM; UPDATE—2008

Despite the fact that viruses have fallen into disfavor as an etiologic paradigm,²⁰ over the past decade, there has been increasing evidence that FCV may have a causative role in the pathogenesis in at least some cases of feline idiopathic cystitis.^{159–161} In the 1980s, transmission electron microscopic examination of urethral matrix- crystalline plugs obtained from male cats with urethral obstruction revealed virus-like particles, similar in size (approximately 25–30 nm) and morphology to caliciviruses (**Fig. 3**).^{8,39} These calicivirus-like particles were observed in 38% of 92 urethral plugs obtained from male cats with urethral obstructive idiopathic cystitis.³⁹ In subsequent studies, an improved virus isolation technique enabled isolation of two new FCV strains (designated FCV-U1 and FCV-U2) from urine obtained from cats with idiopathic cystitis (**Fig. 4**).¹⁶¹ Genetic analyses revealed that both new urinary FCV strains were genetically distinct from vaccine and other wild-type strains.¹⁶¹ The observation of calicivirus-like particles in urethral plugs and isolation of two new FCV strains from urine of cats with idiopathic cystitis led the authors to reexamine the role of caliciviruses in feline idiopathic cystitis.

Although isolation of FCV from the urinary tract of affected cats is an essential prerequisite to establishing a “cause and effect” relationship of FCV with idiopathic cystitis, isolation does not necessarily imply causation. Koch’s postulates have historically been used to establish a causal relationship between a microorganism and a disease.¹⁶² In recognition of the unique biology of viruses and technological advances in virus identification, however, other investigators have introduced modifications to Koch’s postulates.¹⁶³ Based on the unified concept of causation proposed by Evans,¹⁶³ criteria for establishing a causative relationship between FCV and idiopathic cystitis should include the following: 1) prevalence of the disease should be significantly higher in cats exposed to FCV than in cats not exposed; 2) FCV should be more common in cats with the disease than in cats without the disease when all risk factors are held constant; 3) temporally, the disease should follow exposure to FCV with a distribution of incubation periods on a bell-shaped curve; 4) a spectrum of host responses should follow exposure to FCV along a biological gradient from mild to severe; 5) a measurable host response following exposure to FCV should regularly appear in cats lacking previous exposure or increase in magnitude if previous

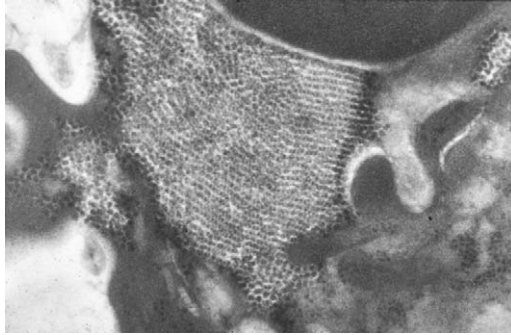


Fig. 3. Transmission electron photomicrograph of a section of matrix-crystalline urethral plug removed from a 3-year-old domestic long-haired cat. Note viruslike particles contained in an unidentified cell (original magnification $\times 25,600$).

exposure occurred; 6) the frequency of induced disease should be higher in cats appropriately exposed to FCV than in those not so exposed; 7) elimination or modification of FCV should decrease the incidence of the disease; and 8) modification of the host's responses on exposure to FCV (eg, effective immunization) should decrease or eliminate the disease.

Large-scale epidemiologic and experimental studies necessary to establish causation have been hindered by lack of a sensitive and rapid means of detecting FCV

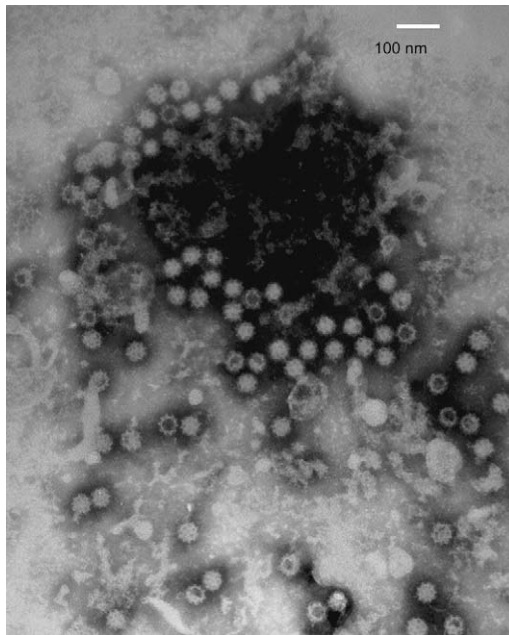


Fig. 4. Transmission electron photomicrograph of feline calicivirus (strain FCV-U2) particles isolated by cell-culture inoculation of urine obtained from a 3-year-old castrated male cat with obstructive idiopathic lower urinary tract disease. Negatively stained with phosphotungstic acid (bar = 100 nm; original magnification $\times 73,000$). (Photo courtesy of Dan Taylor, Michigan State University Diagnostic Center for Population and Animal Health.)

urinary tract infections. Because clinical signs, clinical laboratory data, and microscopic evaluation of tissues stained for routine light microscopy cannot reliably distinguish FCV-induced disease from other causes of lower urinary tract signs, the authors developed a FCV p30 gene-based real-time reverse transcription-polymerase chain reaction (RT-PCR) assay optimized for feline urine that is capable of detecting low virus concentrations in urine and a broad range of FCV isolates.^{43,44} This RT-PCR assay along with conventional light microscopic methods were used to characterize lower urinary tract disease induced by infection with urinary and respiratory strains of FCV in specific-pathogen free (SPF) cats.¹⁵⁹ FCV was detected by RT-PCR with similar frequency in urinary tract tissues from cats infected with either the urinary or respiratory strain of FCV, and immuno-histochemical staining confirmed localization of FCV antigen to the bladder urothelium. Light microscopy revealed that urinary bladder lesions were more severe and more frequent in cats infected with the urinary strain; oral/respiratory tract lesions were more severe and more frequent in cats infected with the respiratory strain.

In addition, the authors investigated the prevalence of FCV urinary tract infection and FCV antibodies in cats with and without idiopathic cystitis using the FCV RT-PCR assay and a virus neutralizing antibody assay.¹⁶⁰ Study populations included client-owned cats with nonobstructive or obstructive idiopathic cystitis, client-owned or sheltered cats with FCV-induced upper respiratory tract disease, and client-owned or colony-housed healthy asymptomatic cats. FCV nucleic acids were detected with similar frequency (approximately 6%) in urine from cats with nonobstructive or obstructive idiopathic cystitis and cats with upper respiratory tract disease; FCV was not detected in urine from asymptomatic cats. FCV was detected in urine but not in oropharyngeal secretions from three cats with idiopathic cystitis. Mean FCV neutralizing antibody titers for cats with nonobstructive or obstructive idiopathic cystitis and upper respiratory tract disease were significantly higher than the mean titer of asymptomatic control cats. These results confirm urinary shedding of FCV in cats with and without respiratory signs and suggest increased exposure to FCV in cats with idiopathic cystitis compared with asymptomatic controls.

These observations are consistent with the hypothesis that FCV has a causative role in the pathogenesis of at least some cases of feline idiopathic cystitis. Symptomatic disease induced by an infectious agent is the cumulative result of a microorganism's ability to establish infection and compromise host function and, conversely, of a host's ability to resist or curtail infection.¹⁶⁴ Because the virulence of a specific viral uropathogen and the resistance of a given host may vary considerably, FCV urinary tract infections may be asymptomatic or may be associated with substantial morbidity. In general, viral uropathogens may cause disease by: (1) inducing cell injury or death, (2) altering cellular functions, (3) producing toxic compounds or waste products, and (4) stimulating or suppressing an immune response.¹⁶⁴⁻¹⁶⁶ Recent studies have identified the components of the functional receptor for FCV and include the junctional adhesion molecule 1 (JAM-1) and alpha 2,6-linked sialic acid.^{167,168} JAM-1 is specifically localized to tight junctions of epithelial and endothelial cells and is involved in regulation of tight junction integrity and permeability, and leukocyte-endothelial cell interactions.¹⁶⁹ Sialic acids are negatively charged sugar molecules usually located at the ends of oligosaccharides attached to glycoproteins, glycolipids, and proteoglycans on the cell surface.¹⁶⁸ It is likely that JAM-1 and sialic acid serve as serotype-independent tissue receptor components capable of mediating virus attachment, infection, and intracellular signaling.^{167,168,170} Variations in host receptors may govern host-virus interactions and may play important roles in tissue tropism, organ pathogenesis, and variations observed in host phenotypic expression of infection.

Because urothelial cell tight junctions and GAG layer are integral components of the urothelial barrier, the authors speculate that FCV attachment, infection, and induction of cellular injury or death may alter the structural and/or functional integrity of the urothelium. Subsequent exposure of bladder wall tissues to urine constituents could result in sensory afferent nerve stimulation, mast cell activation, and/or induction of immune-mediated or neurogenic inflammatory responses.^{10,73,111} Studies to define the causative role of FCV in the pathogenesis of idiopathic cystitis are in progress.

Summary and Caveats for Future Studies

Since 1996 when the authors assembled the existing body of knowledge about lower urinary tract disease in cats, the understanding of the biological behavior and pathologic features of naturally occurring feline idiopathic cystitis has increased. This increase in knowledge is attributed to results of appropriately designed clinical studies, recognition that laboratory models may provide useful or misleading information, and recognition of the limitations of extrapolating information derived from other species. It is clear that no single model explains all the biological variability observed in cats with idiopathic cystitis. The authors' experience and available evidence indicate that feline idiopathic cystitis represents a syndrome resulting from a number of separate underlying but potentially interrelated mechanisms rather than a disease with a single cause. Identification of safe and effective treatment and prevention strategies of feline idiopathic cystitis will likely vary, depending on the underlying causes.

We are actively seeking identification of causes, combined with results of appropriately controlled studies designed to evaluate the safety and efficacy specific, supportive, and/or symptomatic modes of therapy. Toward this end, future studies designed to identify causative factors and effective treatment strategies should encompass the following principles:

1. Clinical signs of hematuria, dysuria, and pollakiuria may result from fundamentally different causes. There is no single specific diagnostic test or marker that can be relied upon to detect idiopathic cystitis. For patients enrolled in clinical or epidemiologic studies, other causes of lower urinary tract signs must be excluded by use of contemporary diagnostic methods.
2. Various forms of idiopathic cystitis may be encountered in cats with naturally occurring disease (eg, acute self-limiting, chronic persistent or frequently recurrent, or obstructive). Whether these forms represent extremes in the spectrum of clinical manifestations associated with similar etiologic factors, or whether they represent entirely different mechanisms of disease is unknown. Consequently, results of studies performed in cats with one form of idiopathic cystitis should not necessarily be extrapolated to cats with other forms of the disease.
3. Recurrent clinical signs in patients with idiopathic cystitis are often assumed to be recurrence of the original disease; however, recurrent signs may also be the result of an onset of a different lower urinary tract disease associated with similar clinical signs. This fact should be considered when formulating exclusion and inclusion criteria.
4. Tissues of the lower urinary tract respond to different etiologic agent in a limited number of ways. Meaningful studies must include appropriate control populations to ensure that observed abnormalities are specific for the type of lower urinary tract being studied.
5. Clinical signs subside within 5–7 days without therapy in the majority of cats with idiopathic cystitis. The self-limiting nature of clinical signs in most cats with idiopathic cystitis underscores the need for controlled randomized double-blind

clinical trials rather than anecdotes and opinions when assessing the safety and efficacy of putative therapies.

ILLUSTRATIVE CASE REPORT

A 3 year-old castrated male domestic short hair cat was presented with a 5-day history of acute pollakiuria and dysuria that had progressed to stranguria, vocalizing, and anuria over the past 24 hours. The owner reported no history of any prior illnesses. The cat lived indoors and was fed a 100% dry maintenance diet. Another young female cat had recently been added to the family household. The affected cat was vaccinated for feline calicivirus 24 months ago. Physical examination revealed tachycardia, tachypnea, and a large overdistended painful urinary bladder. Results of a CBC and serum chemical profile were normal. Analysis of a urine specimen collected just before urethral catheterization revealed a urine specific gravity of 1.052, a pH of 7.2 (pH meter), hematuria, proteinuria, and moderate struvite crystalluria. Quantitative culture of urine for aerobic bacteria was negative. Survey abdominal radiographs did not identify any radiopaque uroliths. The cat was anesthetized and urethral patency was restored with a combination of antegrade and retrograde flushing of a crystalline-matrix plug lodged in the distal urethra. Unfortunately, the urethral plug material collected was lost and therefore not available for analysis. Hydration was maintained by intravenous fluid administration and short-term narcotic analgesics were administered. Urethral patency was maintained by placement of a 3.5 French indwelling red rubber urethral catheter. After 24 hours the urinary catheter was removed and empiric antimicrobial therapy was initiated. The cat recovered uneventfully and was discharged 3 days after admission with instructions to feed a struvite calculolytic diet for 30 days. In follow-up studies of samples collected at the time of admission, feline calicivirus nucleic acids were detected by a RT-PCR assay in urine, but not in oropharyngeal secretions.^{43,44} A serum FCV neutralizing antibody titer was 1:4096. Once at home, the owner reported that the cat refused to eat the calculolytic diet and therefore was fed its original dry diet. The cat experienced one episode of recurrence of signs of lower urinary tract disease approximately two years later.

Urethral obstruction in cats with idiopathic cystitis may result from: (1) inflammatory swelling of the urethra, (2) urethral muscular spasm, (3) reflex dyssynergia, (4) intraluminal accumulations of sloughed tissue, inflammatory cells, or red blood cells or (5) formation of matrix-crystalline urethral plugs.¹⁸ Physical evidence suggests that formation of a crystalline-matrix urethral plug as sequelae of idiopathic cystitis was responsible, at least in part, for the obstructive uropathy observed in this case. The authors have hypothesized that formation of matrix-crystalline urethral plugs in cats with idiopathic cystitis occur as a result of cystitis-induced increased inflammatory matrix production in conjunction with a concomitant, but etiologically unrelated, crystalluria.¹⁷¹ Supporting this hypothesis is the observation of recurrent episodes of non-obstructive hematuria and dysuria, and episodes of plug-induced urethral obstruction in other cats with idiopathic cystitis.¹⁸ These observations suggest that male cats with idiopathic cystitis and concomitant crystalluria are at risk for formation of matrix-crystalline urethral plugs and urethral obstruction. The biologic behavior of this case provides evidence that support use of acidifying magnesium-restricted diets to minimize recurrence of matrix-crystalline urethral plugs containing struvite.

Several causative factors may be involved in the pathogenesis of idiopathic cystitis in this case. Introduction of an additional cat into the household could have been a psychological stressor of sufficient magnitude to induce central dysregulation of autonomic neurons regulating bladder contraction and/or directly induced urothelial

injury and subsequent signs and sequelae of idiopathic cystitis. Alternatively, detection of FCV in urine and high antibody titer in serum are indicative of an active viral urinary tract infection and suggest that FCV may be causatively involved, either by direct urothelial injury or indirectly as a trigger of neurogenic inflammation or psychoneuroendocrine dysfunction. The authors have detected FCV viruria in a number of cats with idiopathic cystitis without concurrent signs of upper respiratory tract disease or evidence of oral virus shedding.^{160,161} Disparities in disease phenotype, tissue tropism, and virus-shedding patterns between FCV strains may be the result of variation in virus virulence, in expression and distribution of host tissue receptors, or in host defense mechanisms. Regardless, detection of FCV viruria and a dramatic concomitant host antibody response support the notion that FCV may be a causative agent in some cases of idiopathic cystitis and emphasize the need for further studies characterizing a cause-and-effect relationship of FCV with idiopathic cystitis.

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